20251110EB_MN (vLH78, 51, mitoSOX)

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2025年11月10日 星期一

soma_vLH51_MN_100x_1xzoom_001split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs

2025年11月12日 星期三

T2 -> T3

2025年11月14日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年11月17日 星期一

T4 -> T5

2025年11月19日 星期三

T5 -> T6

2025年11月21日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年11月24日 星期一

T6 -> T7
Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.

9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info

	A	B
1	cell	passage
2	WT11	

Table1

	A	B
1	run1	volume (mL)
2	trypsin	3
3	HI-FBS/CTWM	6
4	run2	*take some undissolved EB
5	trypsin	2
6	HI-FBS/CTWM	4

Table3

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	4.26	99	7	29.82	A	96	60	36	507
3					B	24	250	16	939
4					C,D	6	1972	12	the rest

Plating

<table>
<tr><th></th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr>
<tr><td>A</td><td></td><td></td><td></td><td rowspan="4">51</td><td rowspan="4">78</td><td rowspan="2">no virus</td></tr>
<tr><td>B</td><td></td><td></td><td></td></tr>
<tr><td>C</td><td></td><td></td><td></td><td></td></tr>
<tr><td>D</td><td></td><td></td><td></td><td></td></tr>
</table>

2025年11月25日 星期二

Change T7 -> T7 if added virus

2025年11月26日 星期三

T7-> T7 if not added virus

2025年11月27日 星期四

T7 -> T8

2025年11月28日 星期五

1/2 T8 -> 1/2 T8

2025年12月1日 星期一

1/2 T8 -> 1/2 T8

2025年12月3日 星期三

1/2 T8 -> 1/2 T8

2025年12月4日 星期四

Treat 24h time point

Well1

	1	2	3	4	5	6
A				1	1	1
B				2	2	2
C				3	3	
D				4	4	

Table5

	A	B	C	D	E	F	G	H	I	J	K	L
1			vLH51				vLH78			mitoSOX		
2	condition#	condition	treat time	image time	soma	neurite	treat time	image time	total position	treat time	mitosox incubation	image time
3	1	2.5mM glucose 1h	10:40	11:40	1-15	1-10	14:00	15:05	20	15:10	16:10	16:43,17:08
4	2	0.15mM glucose 1h	11:20	12:20	16-30	11-20	14:30	15:30	20	15:10	16:10	16:56
5	3	2.5mM glucose 24h	13:00 12/4	13:10	31-45	21-30	13:00 12/4	14:16	20			
6	4	0.15mM glucose 24h	13:00 12/4	13:30	46-60	31-40	13:00 12/4	14:30	20			

vLH51_fileid-conditions

	A	B	C	D
1	image	treatment	treat_time	type
2	soma_vLH51_MN_100x_1xzoom_001	2.5mM gluc	1h	movie
3	soma_vLH51_MN_100x_1xzoom_002	2.5mM gluc	1h	movie
4	soma_vLH51_MN_100x_1xzoom_003	2.5mM gluc	1h	movie
5	soma_vLH51_MN_100x_1xzoom_004	2.5mM gluc	1h	movie
6	soma_vLH51_MN_100x_1xzoom_005	2.5mM gluc	1h	movie
7	soma_vLH51_MN_100x_1xzoom_006	2.5mM gluc	1h	movie
8	soma_vLH51_MN_100x_1xzoom_007	2.5mM gluc	1h	movie
9	soma_vLH51_MN_100x_1xzoom_008	2.5mM gluc	1h	movie
10	soma_vLH51_MN_100x_1xzoom_009	2.5mM gluc	1h	movie
11	soma_vLH51_MN_100x_1xzoom_010	2.5mM gluc	1h	movie
12	soma_vLH51_MN_100x_1xzoom_011	2.5mM gluc	1h	movie
13	soma_vLH51_MN_100x_1xzoom_012	2.5mM gluc	1h	movie
14	soma_vLH51_MN_100x_1xzoom_013	2.5mM gluc	1h	movie
15	soma_vLH51_MN_100x_1xzoom_014	2.5mM gluc	1h	movie
16	soma_vLH51_MN_100x_1xzoom_015	2.5mM gluc	1h	movie
17	soma_vLH51_MN_100x_1xzoom_016	0.15mM gluc	1h	movie
18	soma_vLH51_MN_100x_1xzoom_017	0.15mM gluc	1h	movie
19	soma_vLH51_MN_100x_1xzoom_018	0.15mM gluc	1h	movie
20	soma_vLH51_MN_100x_1xzoom_019	0.15mM gluc	1h	movie
21	soma_vLH51_MN_100x_1xzoom_020	0.15mM gluc	1h	movie
22	soma_vLH51_MN_100x_1xzoom_021	0.15mM gluc	1h	movie
23	soma_vLH51_MN_100x_1xzoom_022	0.15mM gluc	1h	movie
24	soma_vLH51_MN_100x_1xzoom_023	0.15mM gluc	1h	movie
25	soma_vLH51_MN_100x_1xzoom_024	0.15mM gluc	1h	movie
26	soma_vLH51_MN_100x_1xzoom_025	0.15mM gluc	1h	movie
27	soma_vLH51_MN_100x_1xzoom_026	0.15mM gluc	1h	movie
28	soma_vLH51_MN_100x_1xzoom_027	0.15mM gluc	1h	movie
29	soma_vLH51_MN_100x_1xzoom_028	0.15mM gluc	1h	movie
30	soma_vLH51_MN_100x_1xzoom_029	0.15mM gluc	1h	movie
31	soma_vLH51_MN_100x_1xzoom_030	0.15mM gluc	1h	movie
32	soma_vLH51_MN_100x_1xzoom_031	2.5mM gluc	24h	movie
33	soma_vLH51_MN_100x_1xzoom_032	2.5mM gluc	24h	movie

34	soma_vLH51_MN_100x_1xzoom_033	2.5mM gluc	24h	movie
35	soma_vLH51_MN_100x_1xzoom_034	2.5mM gluc	24h	movie
36	soma_vLH51_MN_100x_1xzoom_035	2.5mM gluc	24h	movie
37	soma_vLH51_MN_100x_1xzoom_036	2.5mM gluc	24h	movie
38	soma_vLH51_MN_100x_1xzoom_037	2.5mM gluc	24h	movie
39	soma_vLH51_MN_100x_1xzoom_038	2.5mM gluc	24h	movie
40	soma_vLH51_MN_100x_1xzoom_039	2.5mM gluc	24h	movie
41	soma_vLH51_MN_100x_1xzoom_040	2.5mM gluc	24h	movie
42	soma_vLH51_MN_100x_1xzoom_041	2.5mM gluc	24h	movie
43	soma_vLH51_MN_100x_1xzoom_042	2.5mM gluc	24h	movie
44	soma_vLH51_MN_100x_1xzoom_043	2.5mM gluc	24h	movie
45	soma_vLH51_MN_100x_1xzoom_044	2.5mM gluc	24h	movie
46	soma_vLH51_MN_100x_1xzoom_045	2.5mM gluc	24h	movie
47	soma_vLH51_MN_100x_1xzoom_046	0.15mM gluc	24h	movie
48	soma_vLH51_MN_100x_1xzoom_047	0.15mM gluc	24h	movie
49	soma_vLH51_MN_100x_1xzoom_048	0.15mM gluc	24h	movie
50	soma_vLH51_MN_100x_1xzoom_049	0.15mM gluc	24h	movie
51	soma_vLH51_MN_100x_1xzoom_050	0.15mM gluc	24h	movie
52	soma_vLH51_MN_100x_1xzoom_051	0.15mM gluc	24h	movie
53	soma_vLH51_MN_100x_1xzoom_052	0.15mM gluc	24h	movie
54	soma_vLH51_MN_100x_1xzoom_053	0.15mM gluc	24h	movie
55	soma_vLH51_MN_100x_1xzoom_054	0.15mM gluc	24h	movie
56	soma_vLH51_MN_100x_1xzoom_055	0.15mM gluc	24h	movie
57	soma_vLH51_MN_100x_1xzoom_056	0.15mM gluc	24h	movie
58	soma_vLH51_MN_100x_1xzoom_057	0.15mM gluc	24h	movie
59	soma_vLH51_MN_100x_1xzoom_058	0.15mM gluc	24h	movie
60	soma_vLH51_MN_100x_1xzoom_059	0.15mM gluc	24h	movie
61	soma_vLH51_MN_100x_1xzoom_060	0.15mM gluc	24h	movie
62	neurite_vLH51_MN_100x_1xzoom_001	2.5mM gluc	1h	movie
63	neurite_vLH51_MN_100x_1xzoom_002	2.5mM gluc	1h	movie
64	neurite_vLH51_MN_100x_1xzoom_003	2.5mM gluc	1h	movie
65	neurite_vLH51_MN_100x_1xzoom_004	2.5mM gluc	1h	movie
66	neurite_vLH51_MN_100x_1xzoom_005	2.5mM gluc	1h	movie
67	neurite_vLH51_MN_100x_1xzoom_006	2.5mM gluc	1h	movie
68	neurite_vLH51_MN_100x_1xzoom_007	2.5mM gluc	1h	movie
69	neurite_vLH51_MN_100x_1xzoom_008	2.5mM gluc	1h	movie
70	neurite_vLH51_MN_100x_1xzoom_009	2.5mM gluc	1h	movie

71	neurite_vLH51_MN_100x_1xzoom_010	2.5mM gluc	1h	movie
72	neurite_vLH51_MN_100x_1xzoom_011	0.15mM gluc	1h	movie
73	neurite_vLH51_MN_100x_1xzoom_012	0.15mM gluc	1h	movie
74	neurite_vLH51_MN_100x_1xzoom_013	0.15mM gluc	1h	movie
75	neurite_vLH51_MN_100x_1xzoom_014	0.15mM gluc	1h	movie
76	neurite_vLH51_MN_100x_1xzoom_015	0.15mM gluc	1h	movie
77	neurite_vLH51_MN_100x_1xzoom_016	0.15mM gluc	1h	movie
78	neurite_vLH51_MN_100x_1xzoom_017	0.15mM gluc	1h	movie
79	neurite_vLH51_MN_100x_1xzoom_018	0.15mM gluc	1h	movie
80	neurite_vLH51_MN_100x_1xzoom_019	0.15mM gluc	1h	movie
81	neurite_vLH51_MN_100x_1xzoom_020	0.15mM gluc	1h	movie
82	neurite_vLH51_MN_100x_1xzoom_021	2.5mM gluc	24h	movie
83	neurite_vLH51_MN_100x_1xzoom_022	2.5mM gluc	24h	movie
84	neurite_vLH51_MN_100x_1xzoom_023	2.5mM gluc	24h	movie
85	neurite_vLH51_MN_100x_1xzoom_024	2.5mM gluc	24h	movie
86	neurite_vLH51_MN_100x_1xzoom_025	2.5mM gluc	24h	movie
87	neurite_vLH51_MN_100x_1xzoom_026	2.5mM gluc	24h	movie
88	neurite_vLH51_MN_100x_1xzoom_027	2.5mM gluc	24h	movie
89	neurite_vLH51_MN_100x_1xzoom_028	2.5mM gluc	24h	movie
90	neurite_vLH51_MN_100x_1xzoom_029	2.5mM gluc	24h	movie
91	neurite_vLH51_MN_100x_1xzoom_030	2.5mM gluc	24h	movie
92	neurite_vLH51_MN_100x_1xzoom_031	0.15mM gluc	24h	movie
93	neurite_vLH51_MN_100x_1xzoom_032	0.15mM gluc	24h	movie
94	neurite_vLH51_MN_100x_1xzoom_033	0.15mM gluc	24h	movie
95	neurite_vLH51_MN_100x_1xzoom_034	0.15mM gluc	24h	movie
96	neurite_vLH51_MN_100x_1xzoom_035	0.15mM gluc	24h	movie
97	neurite_vLH51_MN_100x_1xzoom_036	0.15mM gluc	24h	movie
98	neurite_vLH51_MN_100x_1xzoom_037	0.15mM gluc	24h	movie
99	neurite_vLH51_MN_100x_1xzoom_038	0.15mM gluc	24h	movie
100	neurite_vLH51_MN_100x_1xzoom_039	0.15mM gluc	24h	movie
101	neurite_vLH51_MN_100x_1xzoom_040	0.15mM gluc	24h	movie

fileID_treatment_78

	A	B	C	D
1	image	treatment	treat time	type
2	vLH78_MN_100x_1xzoom_0001	2.5mM gluc	24h	movie
3	vLH78_MN_100x_1xzoom_0002	2.5mM gluc	24h	snapshot
4	vLH78_MN_100x_1xzoom_0003	0.15mM gluc	24h	movie
5	vLH78_MN_100x_1xzoom_0004	0.15mM gluc	24h	snapshot
6	vLH78_MN_100x_1xzoom_0005	2.5mM gluc	1h	movie
7	vLH78_MN_100x_1xzoom_0006	2.5mM gluc	1h	snapshot
8	vLH78_MN_100x_1xzoom_0007	0.15mM gluc	1h	movie
9	vLH78_MN_100x_1xzoom_0008	0.15mM gluc	1h	snapshot

MitoSOX

1. Prepare 500nM MitoSOX red: dilute 0.2 uL 5mM mitoSOX into 2 mL HBSS (warm up)
2. treat A6 and B6 with 2.5mM gluc and 0.15mM gluc of full Neurobasal-A from 15:10 - 16:10
3. treat A6 and B6 with 500uL 500nM MitoSOX red in HBSS for 10 min from 16:10-16:20
4. image A1 and B1 in 30 min. (19:30 - 20:00)

Table4

	A	B	C	D	E
1	560	30%	500ms	16bit	
2	DIC		100ms	12bit	soma_001 doesn't have DIC
3	zstack_soma	10um in total	1um/step		
4	zstack_neurite	5um in total	0.5um/step		

Table2

	A	B	C
1	image	treatment	treat_time
2	soma_mitosox_MN_100x_1xzoom_001	2.5mM gluc	1h
3	neurite_mitosox_MN_100x_1xzoom_001	2.5mM gluc	1h
4	soma_mitosox_MN_100x_1xzoom_002	0.15mM gluc	1h
5	neurite_mitosox_MN_100x_1xzoom_002	0.15mM gluc	1h